Innovation, entrepreneurship and business models 3:

Innovation in small and large firms

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Today

- The lecture will focus on how to organize for innovation in small and large firms.
- We will also explore innovation as a process and the role of collaboration on innovation

Literature:

- *Tim Mazzarol, Sophie Reboud (2020) Entrepreneurship and Innovation: Theory, Practice and Context, 4th ed. Series: Springer Texts in Business and Economics. Chapter 4 and 5*

Innovation in small and large firms

- Do small and large firms differ when it comes to innovation -?
  - Resources
  - Organisation of innovation processes
  - Search of new ideas
  - Collaboration patterns
  - Outcome
  - Type of outcome
  - Manager versus entrepreneur

From lecture 1 you learned that innovation is a process and an outcome

- Innovation is process that involves many **interactions** and **feedbacks** in knowledge creation and use.
- In addition, **innovation is based on a learning process that draws on multiple inputs and requires ongoing problem-solving**.

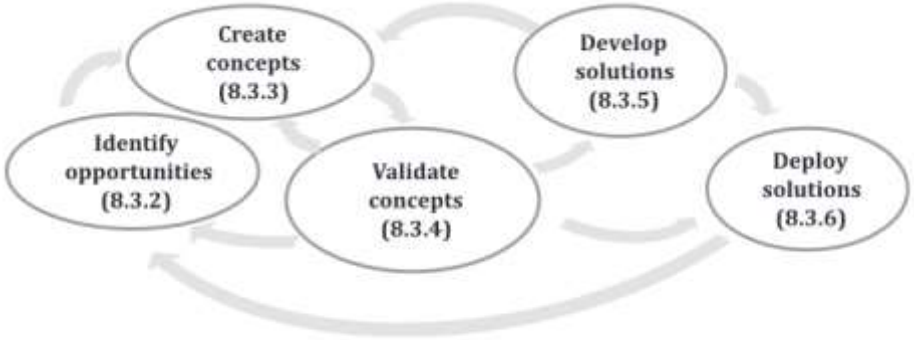


Figure 2 — Innovation processes

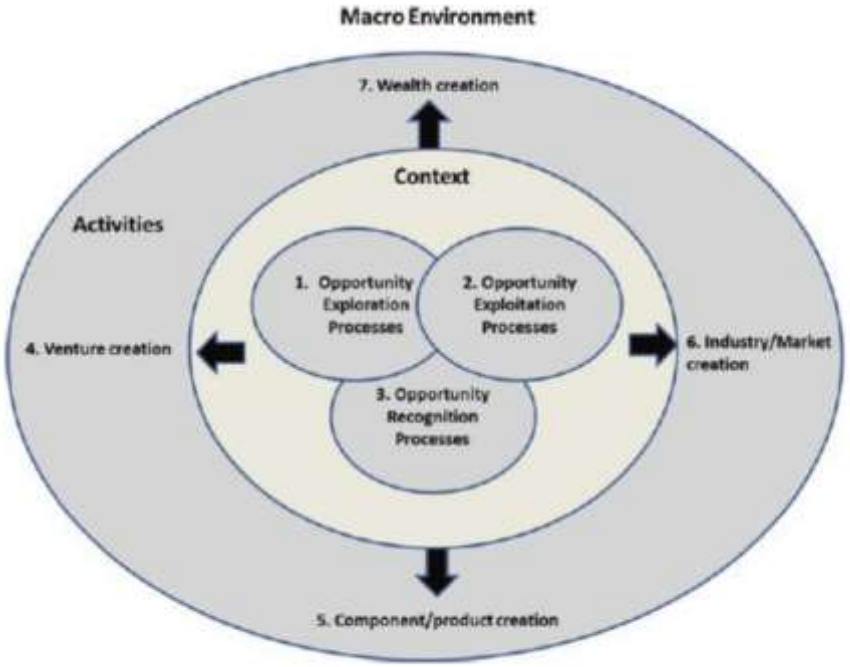


Fig. 1.4 The entrepreneurship domain. (Source: Brush et al. 2003)

Innovation as a non-linear process

Idea → Innovation

time

- Innovation is not a linear, sequential process, but involves many **interactions** and **feedbacks** in knowledge creation and use.
- However, it still makes sense to discuss the different stages of an innovation process

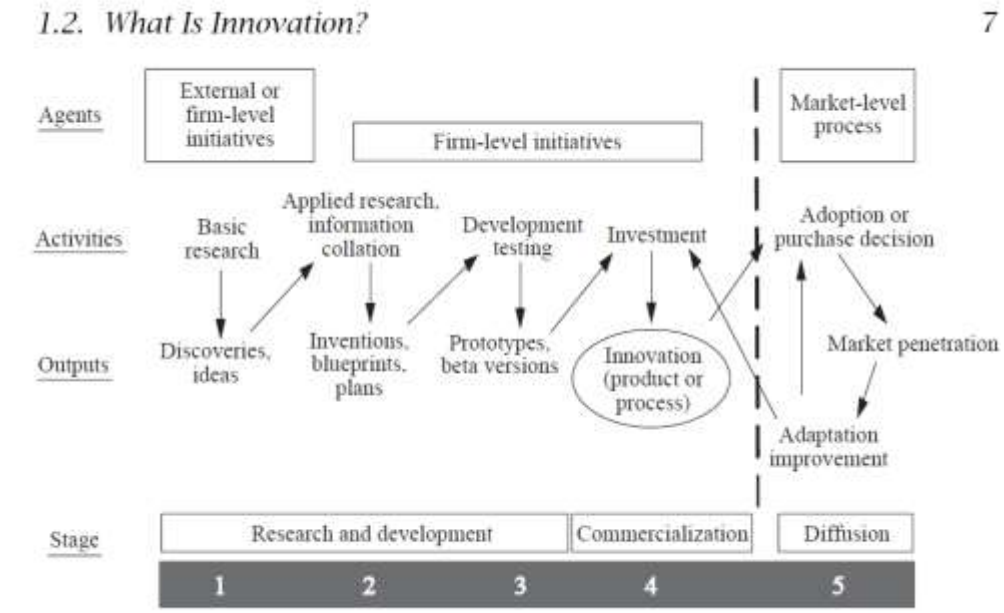
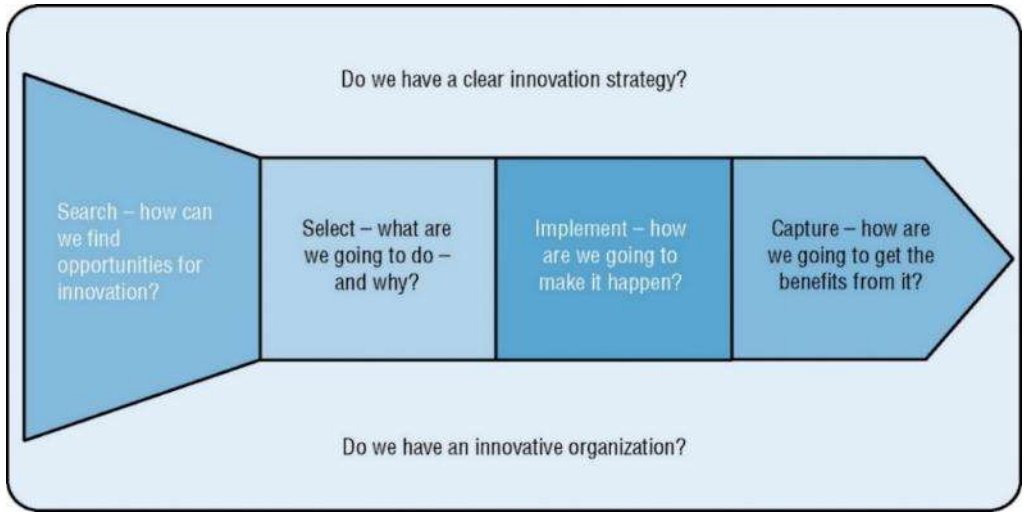


Figure 1.1. The stages of the innovation process.



Understanding innovation: Novelty and impact

Table 1. Innovation and competitive advantage

Type of innovation	Competitive advantage
Disruptive	Re-writing the rules of the competitive game, creating a new 'value proposition'
Radical	Offering a highly novel or unique product or service, premium pricing
Complex	Difficulty of learning about the technology keeps entry barriers high
Continuous incremental innovation	Continuous movement of the cost/performance frontier

Source: Adapted from J. Tidd, J. Bessant and K. Pavitt (2001) *Managing Innovation: Integrating Technological, Market and Organizational Change*, 2nd edition. Chichester: Wiley.

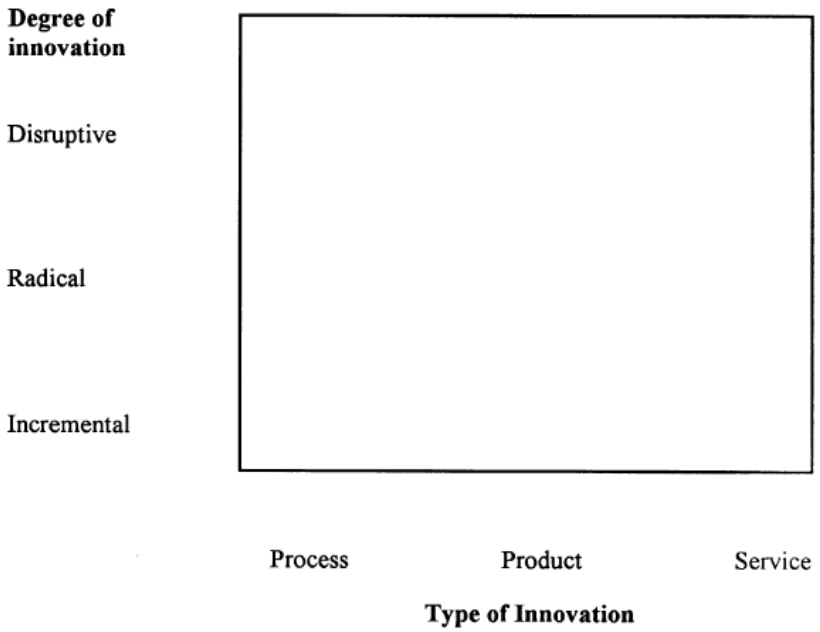


Figure 3. Innovation 'space': the type and degree of innovation.
Source: Adapted from J. Tidd, J. Bessant and K. Pavitt (2001) *Managing Innovation: Integrating*

The growth vector – the type of 'innovation' matters

- Market share growth
- This involves taking the existing products or services to new markets
- New Product Development
- This option involves the firm developing new products or services that can be offered to the same customers or market segment.
- Diversification
- The combination of new products and new markets involves a diversification strategy

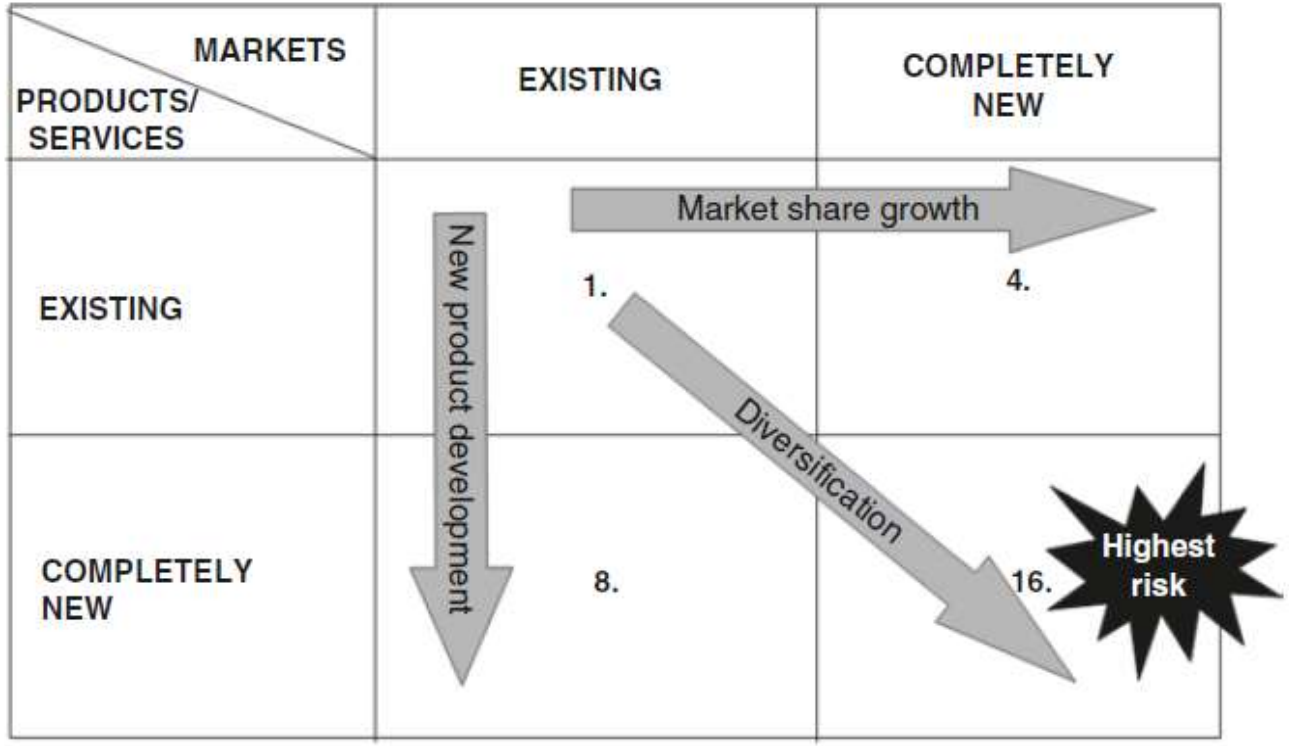
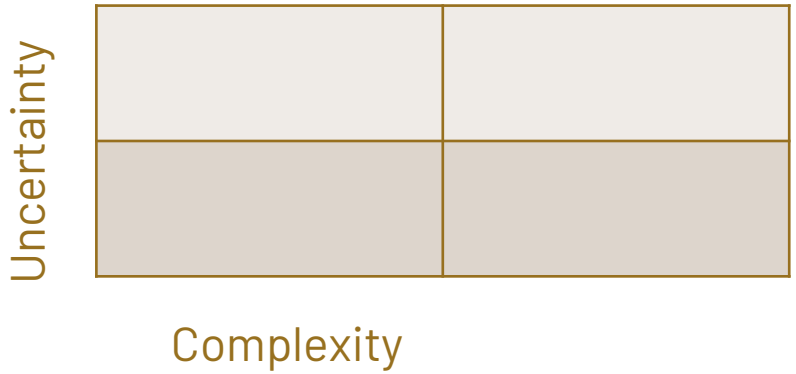


Fig. 5.6 The growth vector framework. (Source: Ansoff 1965)

Innovation is a process defined by uncertainty and complexity

- Uncertainty is a function of the rate of change of technologies and product-market
  - Uncertainty requires e.g., internal R&D
- Complexity is a function of technological and organizational interdependencies
  - Complexity requires e.g., collaboration in networks



Problem

- Where do innovation come from?
- How to find and/or develop a solution to a problem?
- Knowledge – knowledge flows
- Knowledge is non-rival because its use by one organisation or person does not diminish the amount potentially available for use by others (Arrow, 1962). It can spill-over to other organisations on purpose or not.
- Tacit or codified knowledge
- Internal development – use of internal knowledge
  - What do we know how to do
  - What do we normally do
  - How do we use knowledge
- External development- use of external knowledge
  - How to access
  - How to use
  - How to implement

Properties of knowledge

- Need for knowledge
  - How to access the needed knowledge
  - Develop it yourself or collaborate
- Codification of knowledge
  - Tacit vs. codified knowledge
  - Tacit – hard to transfer - requires interaction
  - Codified knowledge – easier to transfer – requires prior knowledge/absorptive capacity
- Different types of knowledge (Johnson and Lundvall, 1994)
  - Know-what
  - Know-why
  - Know-how
  - Know-who

Absorptive Capacity

- **absorptive capacity**, involves the ability of the firm to leverage existing stocks of knowledge, to acquire, absorb and then make effective use of new knowledge. (Cohen and Levinthal, 1990)
- it must possess a culture and structure that encourages and facilitates its employees to possess the necessary mindsets and competencies to acquire, assimilate, transform and exploit knowledge, (Zahra and George 2002).
- This requires not only the firm's ability to effectively connect and communicate with external actors in the exchange of knowledge, but to also connect and communicate with internal actors in this knowledge exchange process (Cohen and Levinthal 1990).

There are many sources of innovation

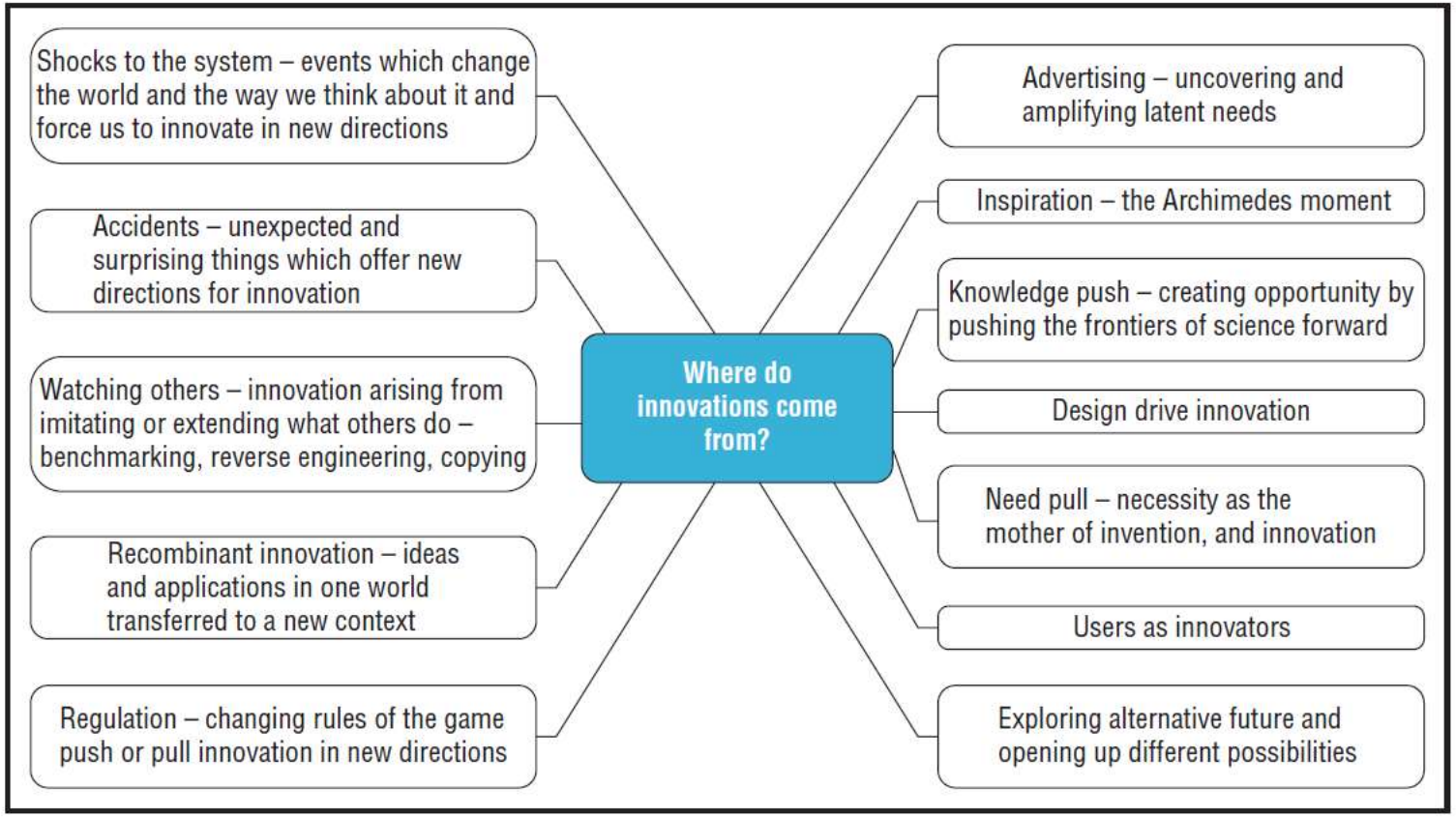


FIGURE 5.1 Where do innovations come from?

Tidd and Bessant 2018

Networking and sources of innovation

- There are multiple sources of innovation
- They are more or less formal and more or less interactive
- Internal
- External
  - Customers, consumers and users
  - Suppliers
  - Competitors
  - Universities and technological institutes
  - Public-Private Partnerships

Open innovation

- Open versus closed innovation model
  - Closed innovation: own R&D proprietary knowledge
  - Open innovation: opening up the organisation's boundaries to a free flow of ideas and knowledge.
    - company's businesses cannot (and should not) wait for the internal technologies to arrive; instead, they should access what they need, as soon as they need it—either from inside the company's own research labs or from the knowledge created in someone else's lab.(Chesbrough 2003)
- Open innovation
  - A key issue for the management of open innovation is the control over intellectual property (IP) rights
  - the manager should approach IP rights not as a protector of their firm's patents, but as a trader in the rights of IP use. This can encompass the licensing of IP rights to other firms, even competitors, to maximise the value capture from the firm's intangible assets.
  - Joint development requires dedicated staff training in the management of IP rights, and well-designed protocols to protect intangible assets and the dissemination of research findings.

Innovation in small and large firms

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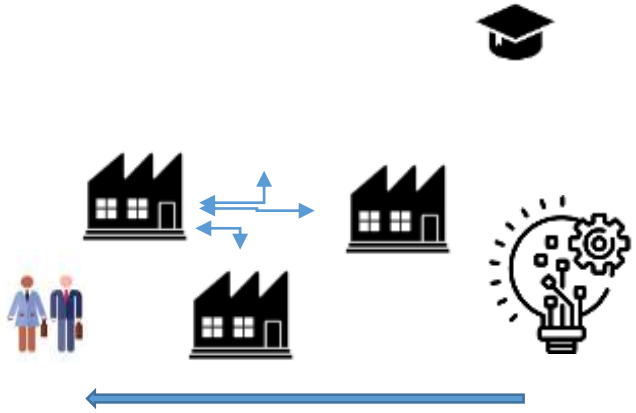
  - Outcome
  - Type of outcome
  - Manager versus entrepreneur



Definition of small, medium, and large firms

Table 5.1 OECD and EU definition of small firms

	Employees	Annual turnover	Assets
Micro-enterprise	1–9	<€2 million	<€2 million
Small enterprise	10–49	<€10 million	<€10 million
Medium-sized enterprise	50–249	<€50 million	<€43 million
Large enterprise	>250	>€50 million	>€43 million

Source: OECD (2004)

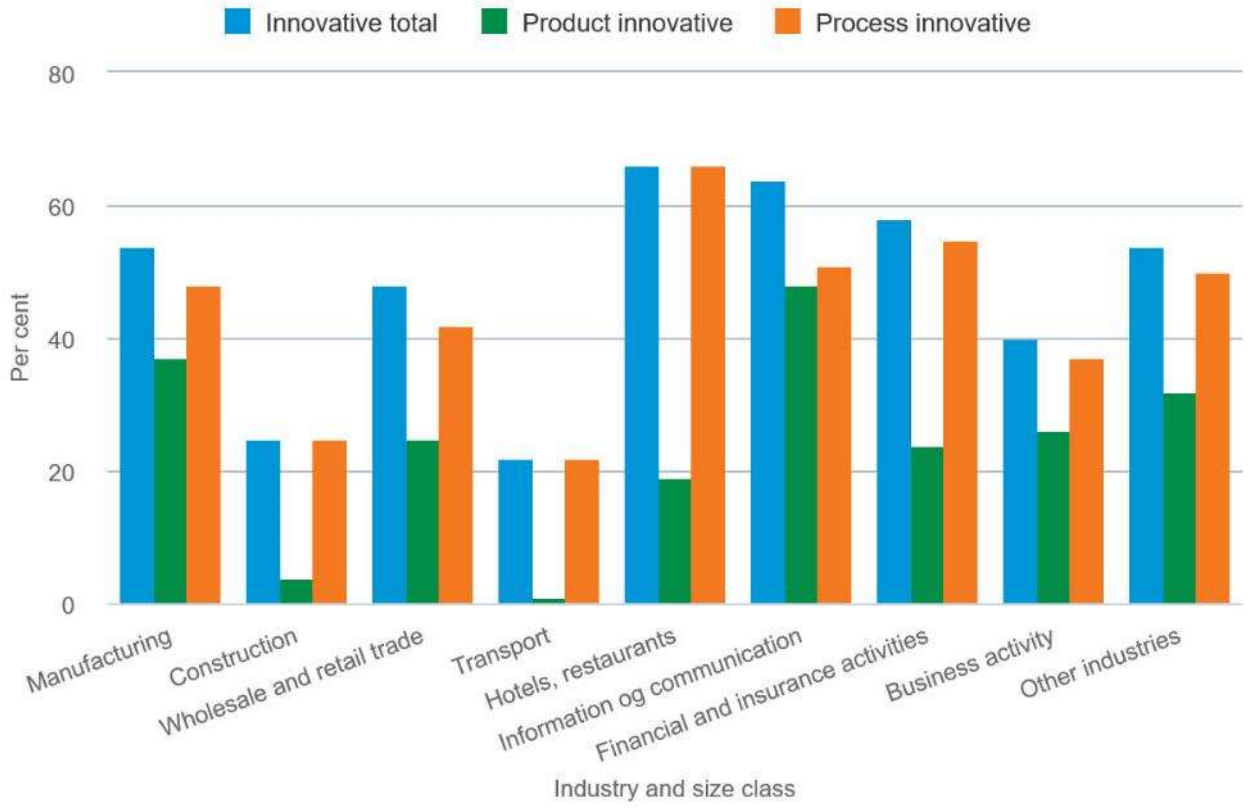
Small and large firms in Denmark

Workplaces and job by industry (DB07 10-grouping), time, size of workplace and unit

	Work places at the end of November	Jobs at the end of November
TOT Industry, total		
2024		
1 job	143 453	143 453
2-4 jobs	72 562	193 228
5-9 jobs	38 021	252 848
10-19 jobs	28 174	384 609
20-49 jobs	21 937	662 762
50-99 jobs	6 963	479 249
100 jobs and more	4 462	1 237 648

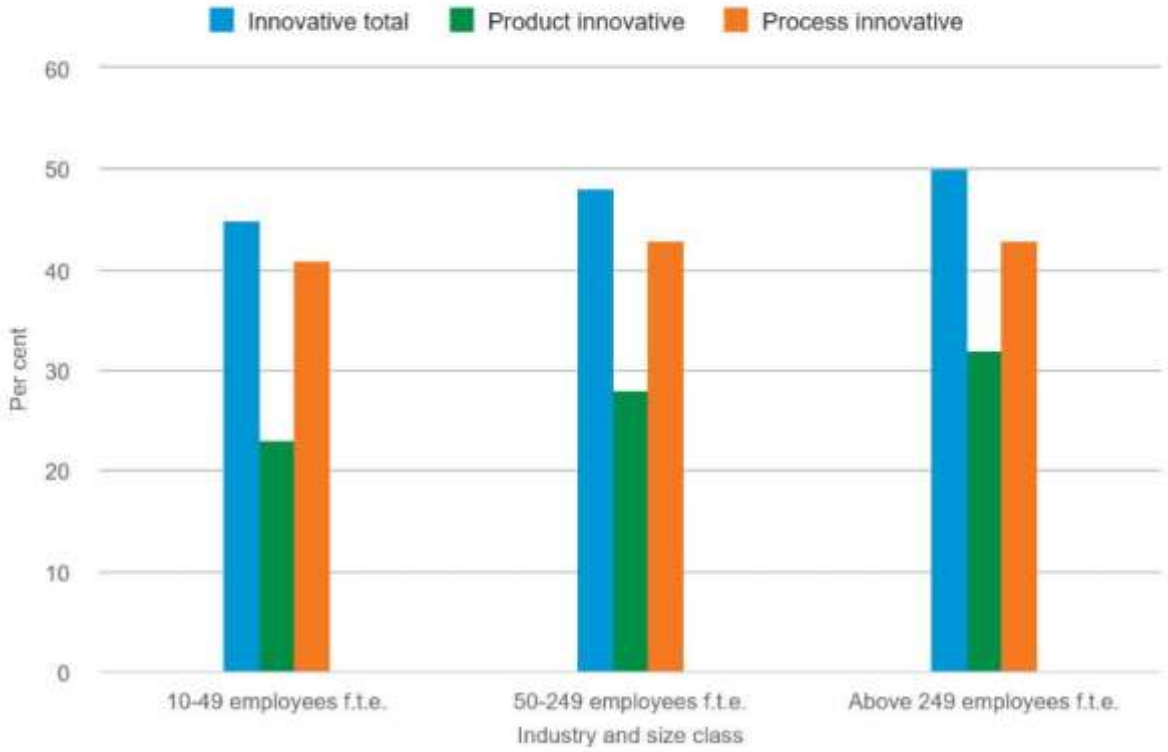
Innovative enterprises

Time: 2022 | Type of innovation:



Innovative enterprises

Time: 2022 | Type of innovation:



Manager – individual - organisation

- Different perspectives to the same 'problem'
- The process of innovation can be viewed from these different perspectives and levels
- All factors can be supportive or restrictive to innovation
- **Remember** the level in the analysis

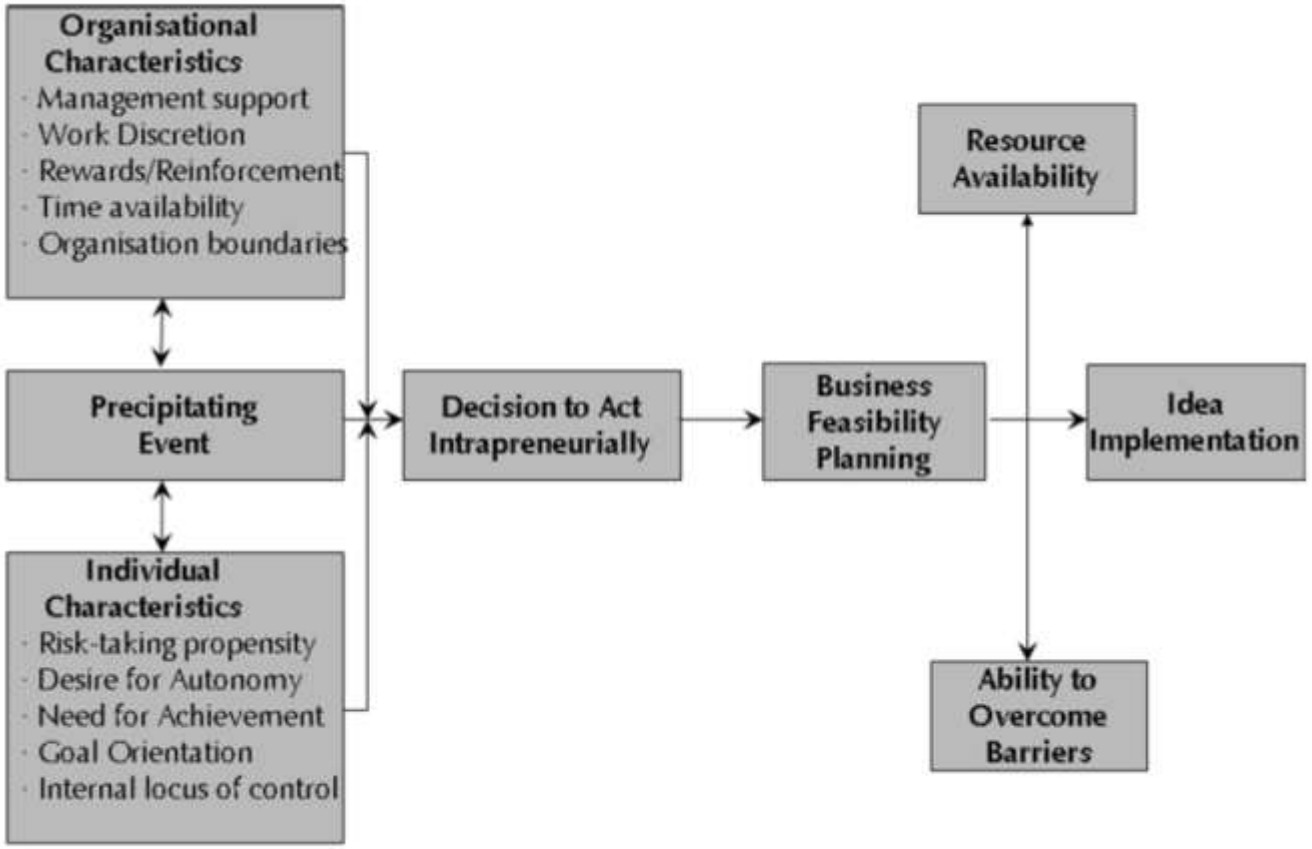


Fig. 4.2 A model of corporate intrapreneuring. (Source: Hornsby et al. 1993)

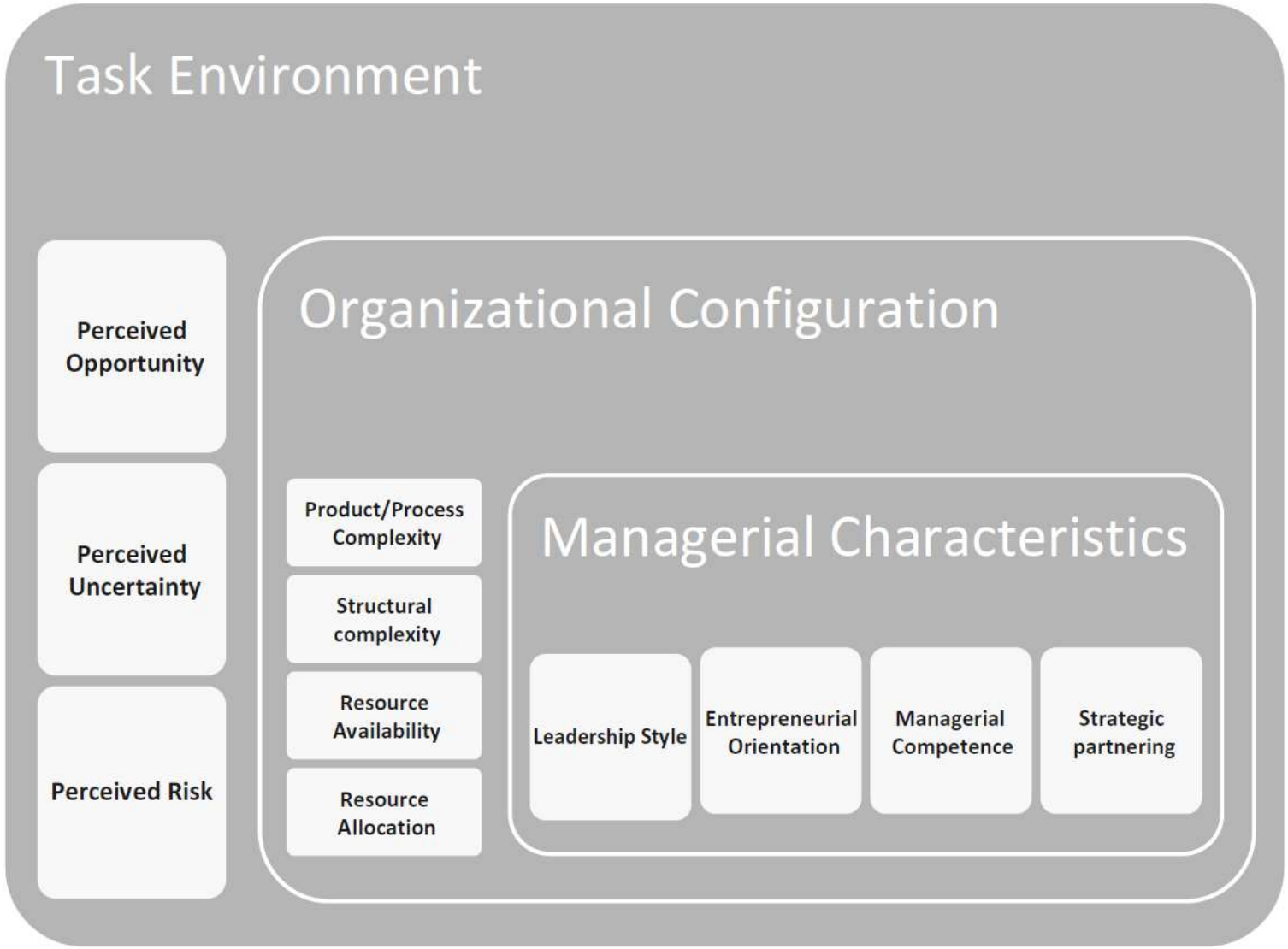


Fig. 5.2 The behaviour of small firms, key units of analysis. (Source: Mazzarol and Reboud 2009)

Strategic management (of innovation)

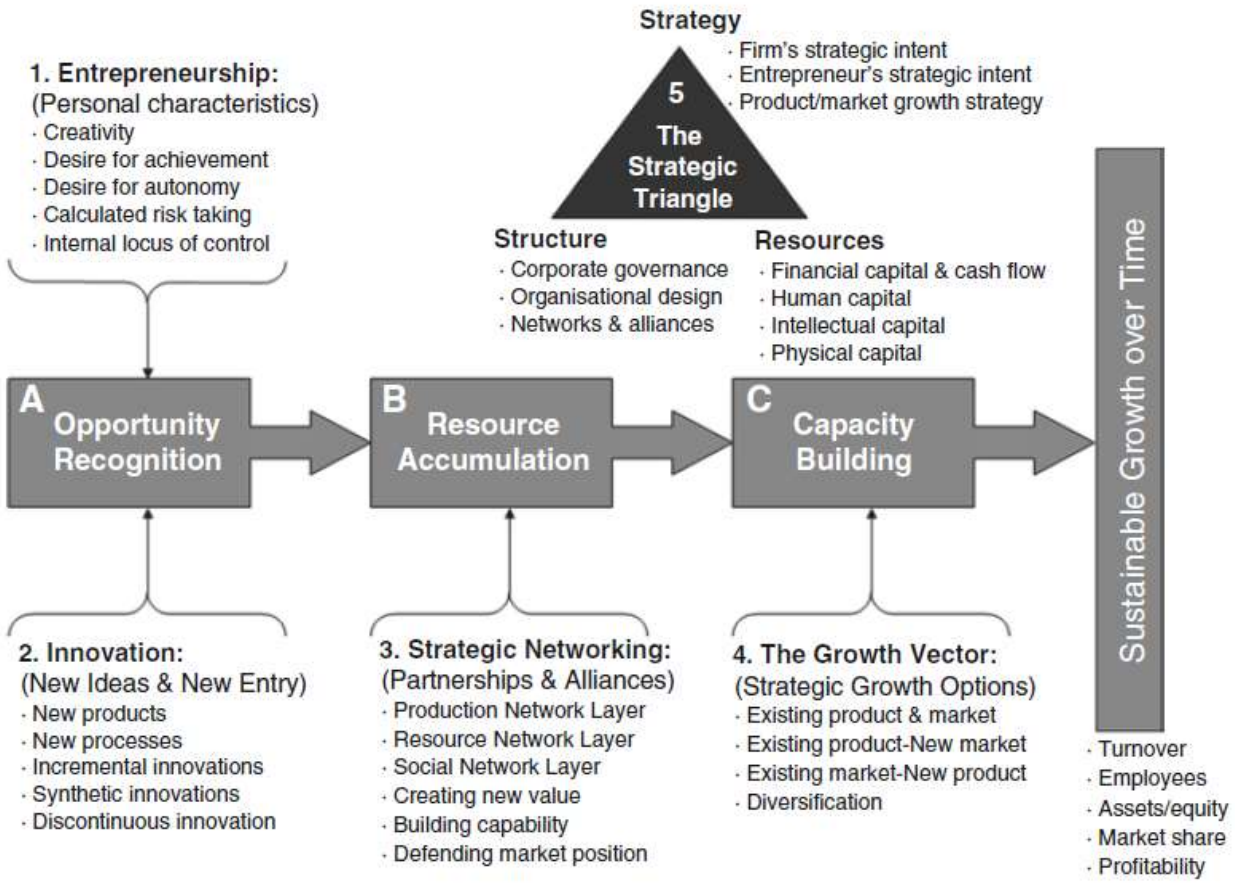


Fig. 5.4 Strategic management of entrepreneurial ventures framework. (Source: Mazzarol 2005)

Strategic management of innovation

- Managing a small firm requires the entrepreneur to keep both strategy and structure in harmony with the firm's resources.

Fig. 5.8 The strategic triangle. (Source: Mazzarol 2005)

Strategic networking

- Potential sources of innovation

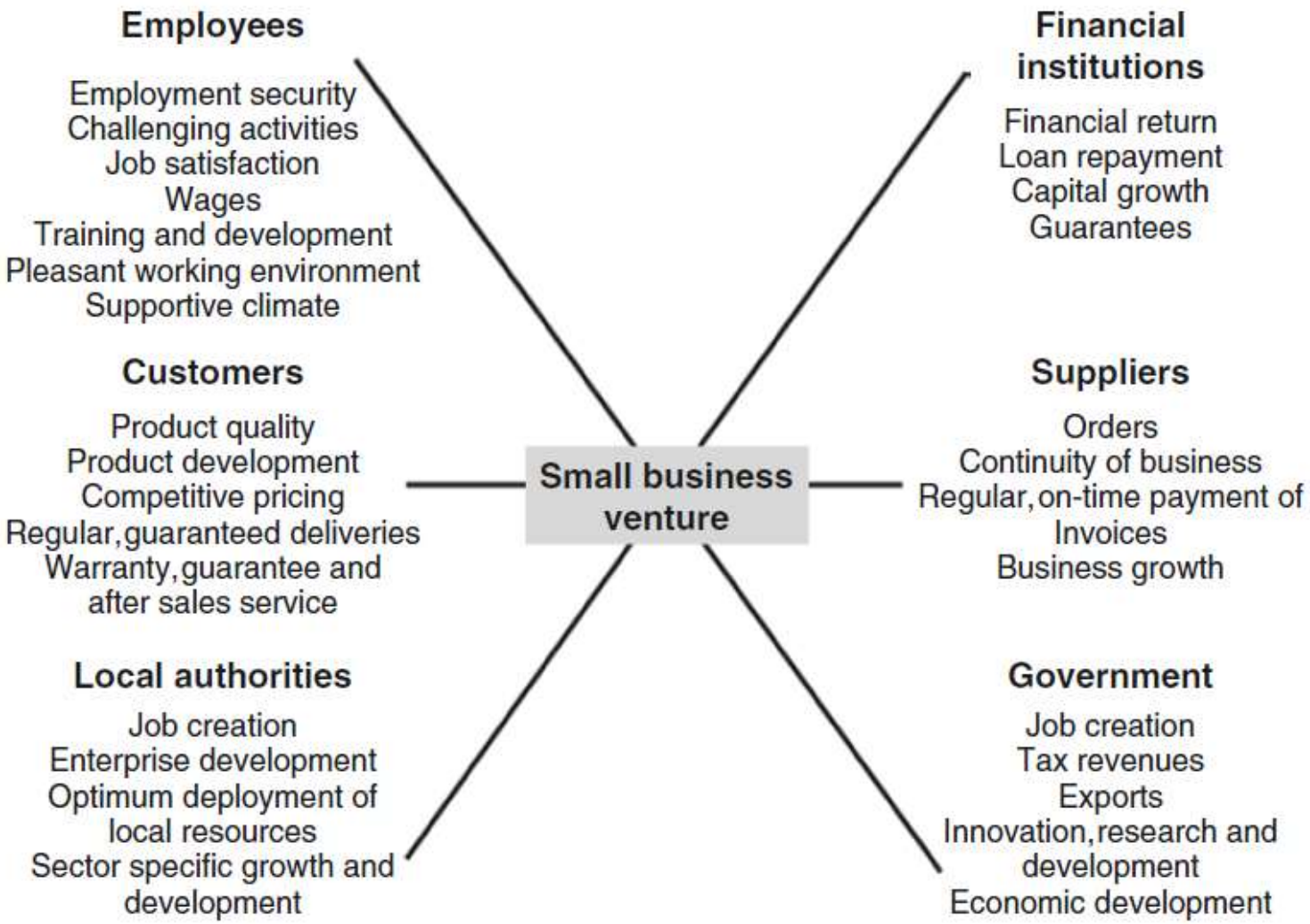


Fig. 5.5 Stakeholder web of the small firm. (Source: Jennings and Beaver 1997)

Forstering innovation

1. The organisational structure should encompass a **strong level of senior management support** for entrepreneurial activity and **innovation** including reward system. Employees should be provided with a high degree of autonomy and discretion over their work
2. the organisation should have fuzzy boundaries that allow **networking between employees and alliance partners from other firms**
3. the human resource strategies should seek to attract and retain achievement-oriented people with the ability to set their own goals and value independence, while having the creativity and risk-taking orientation to try new ideas
4. the senior management should be **tolerant of failure and risk**, but also have the expectation that middle managers will lead innovation as a core part of their job and be provided with appropriate resources

Exercise

- Differences among small and large firm
- Three examples of innovation
  - 1:System Audio
  - 2: Caulipower
  - 3: Novo Nordisk
- Discuss differences
  - Difference in search, select, implement and capture

The small firm 1

- What is the type of innovation?
- How did the firm innovate?
- Where did the idea come from?
- How did the innovation reach the market?

Los Angeles mother-of-two Gail Becker has also been promoting wheat alternatives.

She became frustrated when she could not find healthy alternatives for her sons, who both suffer from coeliac disease, an autoimmune disorder triggered by eating foods containing gluten.

CAULIPOWER-GAIL BECKER

Gail Becker's Caulipower makes cauliflower-based pizza and pasta products

"I tried desperately to find nutritious, gluten-free options that weren't filled with fat, sugar, salt and calories," says Ms Becker. "I stumbled upon recipes for cauliflower crust pizza.

"My creation tasted fine, but it looked awful, my kitchen was a disaster and I had spent 90 minutes I didn't have, as a mum with a full-time job, making a pizza crust! I realised I probably wasn't alone. So I left my job to strike out on my own and create Caulipower."

Caulipower, established in 2016, has since expanded beyond pizza crusts to offer frozen cauliflower pastas. Its products can now be found in more than 25,000 stores across the US.

"Consumers want nutritional food," says Ms Becker. "They no longer expect to have to choose between taste and health, and frankly they shouldn't have to."

The small firm 2

- Ole Witthøft is the owner manager of the high-end speaker company System Audio in Roskilde, which travels around every year and tells customers and retailers about its new products. But when the corona crisis put an end to travel plans, Ole Witthøft and his 13 employees used the time and money they would otherwise spend on travel to develop a smartphone app that the company's customers can use to read the acoustics of a room and adjust the sound of the speaker accordingly.
- "We have never had an app before. It was fantastic to find out: God, we know this, "says Ole Witthøft. The app led to a revenue growth of 42 percent. in 2020 and created attention in the industry globally, which led to the Roskilde company winning the award for the world's best wireless speaker in both 2021 and 2022. The experience with the app has also helped to change the company's mindset in the long run: "We discovered that we may need to rethink our strategy and recognize that it is a digital world we live in and be present digitally with products that people can experiment with at home," says Ole Witthøft, who has now started in the development of yet another app that will strengthen the relationship with customers, even after they have purchased a product.
- https://borsen.dk/nyheder/proselvstaendig/forskere-smver-kan-bruge-corona-laering-i-klimakrisen?utm_source=innovation&utm_medium=row_1&utm_campaign=list_9

- https://system-audio.com/

The large firm

- Novo Nordisk in 2021 announced that the European Medicines Agency's Committee for Medicinal Products for Human Use (CHMP) has adopted a positive opinion, recommending marketing authorisation for Wegovy™ for chronic weight management in adults with obesity.

- Wegovy is based on the same drug as the diabetes drugs Ozempic and Rybelsus, but in a different dose. Has in experiments shown weight loss of approximately 17 percent. on average after 68 weeks of treatment. Now the step 5 study shows that an average weight loss of 15.2 per cent. maintained in 304 patients after two years of treatment.
- Novo Nordisk is aiming for a possible approval for a future obesity based on semaglutide already in the summer of 2021. It appears from Friday's quarterly accounts from the pharmaceutical company, in which it appears, the company has informed the FDA that it wants to redeem a so-called priority review voucher when it around the turn of the year, it is expected to submit an application for a marketing authorization for semaglutide against obesity. A priority review voucher is used to shorten the case processing by the US government from 10 to six months. Novo Nordisk has taken over the voucher from the Australian biotech company Medicines Development for Global Health, which last year received DKK 80 to 100 million from the Danish company.
- *Novo Nordisk is a leading global healthcare company, founded in 1923 and headquartered in Denmark. Our purpose is to drive change to defeat diabetes and other serious chronic diseases such as obesity and rare blood and endocrine disorders. We do so by pioneering scientific breakthroughs, expanding access to our medicines and working to prevent and ultimately cure disease.*

Pipeline overview

DIABETES

Project	Indication	Description	Phase
IcoSema NN1535	T2D[1]	A combination of GLP-1[2] receptor agonist semaglutide and insulin icodec intended for once-weekly subcutaneous treatment.	●● ●●
Icodec NN1436	T1D[3] and T2D	A long-acting basal insulin analogue intended for once-weekly subcutaneous treatment.	●● ●●
CagriSema NN9388	T2D	A combination of amylin analogue cagrilintide and GLP-1 receptor agonist semaglutide intended for once-weekly subcutaneous treatment.	●● ●○
OW GIP[4]/GLP-1 NN9541	T2D	A dual GLP-1/GIP receptor agonist intended for once-weekly subcutaneous treatment.	●● ○○
GELA NN9506	T2D	A collaboration with GE Healthcare, using ultrasound for once-monthly treatment.	●● ○○
Amycretin NN9490	T2D	A unimolecular long-acting GLP-1 and amylin receptor agonist intended for once-daily oral treatment and once-weekly subcutaneous treatment.	●● ○○
Pumpsulin NN1471	T1D	A novel insulin analogue for use in closed loop pump systems.	●○ ○○
DNA immunotherapy NN9041	T1D	A novel plasmid encoding pre-proinsulin, TGF[5]-beta1, IL[6]-10 and IL-2 intended for subcutaneous treatment.	●○ ○○
OW Oral Semaglutide NN9904	T2D	An oral version of the GLP-1 receptor agonist intended for once-weekly treatment.	●○ ○○
GSI[7] NN1644	T1D	An injectable glucose sensitive insulin intended for once daily treatment.	●○ ○○

OBESITY

Project	Indication	Description	Phase
Oral Semaglutide NN9932	Obesity	A long-acting GLP-1 receptor agonist, 25 mg and 50 mg, intended for once-daily oral treatment.	●● ●○
Semaglutide 7.2 mg NN9536	Obesity	A long-acting GLP-1 receptor agonist, 7.2 mg, intended for once-weekly subcutaneous treatment.	●● ●○
CagriSema NN9838	Obesity	A combination of amylin analogue cagrilintide and GLP-1 receptor agonist semaglutide intended for once-weekly subcutaneous treatment.	●● ●○
GELA NN9505	Obesity	A collaboration with GE Healthcare, using ultrasound for once-monthly treatment.	●● ○○
Monlunabant NN9440	Obesity	CB-1[8] receptor inverse agonist intended for once-daily oral treatment.	●● ○○
Cagrilintide NN9833	Obesity	An amylin analogue intended for once-weekly subcutaneous treatment.	●● ○○
Amycretin NN9487	Obesity	A unimolecular long-acting GLP-1 and amylin receptor agonist intended for once-daily oral treatment and once-weekly subcutaneous treatment.	●● ○○
INV-347 NN9441	Obesity	CB-1 receptor inverse agonist intended for once-daily oral treatment.	●○ ○○
OW GIP/GLP-1 NN9542	Obesity	A dual GLP-1/GIP receptor agonist intended for once-weekly subcutaneous treatment.	●● ○○
Triple NN9662	Obesity	Tri-agonist.	●○ ○○
Amylin 355 NN9638	Obesity	Amylin analogue developed for once-weekly subcutaneous treatment.	●○ ○○

RARE DISEASE

Project	Indication	Description	Phase
Mim8 NN7769	Haemophilia A w/wo inhibitors	A next generation FVIIIa mimetic bispecific antibody intended for subcutaneous prophylaxis for haemophilia A.	●● ●○
Etavopivat NN7535	Sickle cell disease	A second-generation small molecule PKR[9]-activator intended for once-daily oral treatment.	●● ●○
Etavopivat NN7536	Thalassemia	A second-generation small molecule PKR-activator intended for once-daily oral treatment.	●● ○○
NDec NN7533	Sickle cell disease	An oral combination of decitabine and tetrahydrouridine. The project is developed in collaboration with EpiDestiny.	●● ○○
TMPRSS2 RNAi[10]	Hereditary haemo-chromatosis	Small interfering RNA intended for once every 1 to 3 months subcutaneous treatment.	●○ ○○
Inno8 NN7441	Haemophilia A w/wo inhibitors	An antibody intended for oral administration.	●○ ○○

CARDIOVASCULAR & EMERGING THERAPY AREAS

Project	Indication	Description	Phase
Ziltivekimab NN6018	CKD[11] ASCVD[12] AMI[13] HFpEF[14]	A once-monthly monoclonal antibody intended for inhibition of IL-6 activity.	●● ●○
Coramitug NN6019	CVD[15]	An anti-amyloid immunotherapy intended for intravenous treatment.	●● ○○
CM4HF NN9003	CVD	An investigational cell therapy intended for restoring heart function in people with chronic heart failure.	●○ ○○
Anti-ANGPTL3 mAb NN6491	CVD	An ANGPTL3[16] neutralising sweeping antibody intended for once-monthly subcutaneous treatment.	●○ ○○
Semaglutide NN6535	Alzheimer's	A long-acting GLP-1 receptor agonist intended for once-daily oral or once-weekly subcutaneous treatment.	●● ●○
Semaglutide NN9931	MASH[17]	A long-acting GLP-1 receptor agonist intended for once-weekly subcutaneous treatment.	●● ●○
CagriSema NN9588	MASH	A combination of amylin analogue cagrilintide and GLP-1 analogue semaglutide intended for once-weekly subcutaneous treatment.	●● ○○
Zalfermin NN9500	MASH	A long-acting FGF21[18] analogue intended for once-weekly subcutaneous treatment.	●● ○○
CDR132L NN6706	Heart failure	An RNA[19]-based oligonucleotide inhibitor developed for once-monthly intravenous treatment.	●● ○○
LXRa[20] NN6582	MASH	A siRNA[21] targeting LXRa intended for once-monthly subcutaneous treatment.	●○ ○○
MARC1[22] NN6581	MASH	A siRNA molecule targeting MARC1 intended for once-monthly subcutaneous treatment.	●○ ○○
SC4PD NN9001	Parkinson's	Cryopreserved cell therapy developed for disease modifying treatment.	●○ ○○
DCR-XDH NN4004	Gout	An RNA-based oligonucleotide intended for subcutaneous treatment.	●○ ○○
Ventus NLRP3i[23] NN6022	CVD	Small molecule NLRP3 inhibitor intended for once-daily oral treatment.	●○ ○○
CNP HF NN6537	Heart failure	C-type natriuretic peptide intended for once-weekly subcutaneous treatment.	●○ ○○
PD-L1[24] NN4003	Oncology	A PD-L1 GalXC™-derived lipid conjugate intended for once-monthly subcutaneous treatment.	●○ ○○
STAT3 NN4002	Oncology	A GalXC™-derived lipid conjugate one-time subcutaneous treatment.	●○ ○○

● Status in 2023 ● Progress in 2024

○○ Phase 1 ○○ Phase 2 ○○ Phase 3 ○○ Submission and/or approval

1. T2D: Type 2 diabetes. 2. GLP-1: Glucagon-like peptide-1. 3. T1D: Type 1 diabetes. 4. GIP: Gastric inhibitory polypeptide. 5. TGF: Transforming growth factor. 6. IL: Interleukin. 7. GSI: Glucose-sensitive insulin. 8. CB-1: Cannabinoid receptor-1. 9. PKR: Pyruvate kinase-R. 10. RNAi: Ribonucleic acid interference. 11. CKD: Chronic kidney disease. 12. ASCVD: Atherosclerotic cardiovascular disease. 13. AMI: Acute miocardial infarction. 14. HFpEF: Heart failure with preserved ejection fraction. 15. CVD: Cardiovascular disease. 16. ANGPTL3: Angiopoietin-like 3. 17. MASH: Metabolic dysfunction-associated steatohepatitis. 18. FGF21: Fibroblast growth factor 21. 19. RNA: Ribonucleic acid. 20. LXRa: Liver X receptor alpha. 21. siRNA: Small interfering RNA. 22. MARC1: Mitochondrial amidoxime-reducing component 1. 23. NLRP3i: NOD-like receptor protein 3 inhibitor. 24. PD-L1: Programmed death ligand 1.

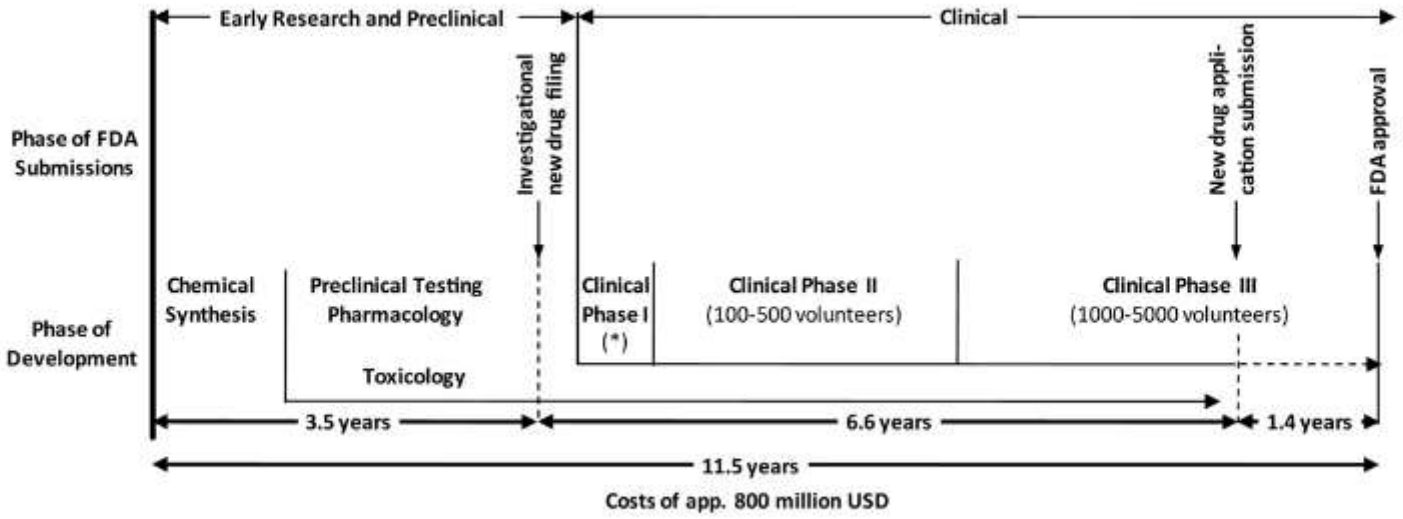


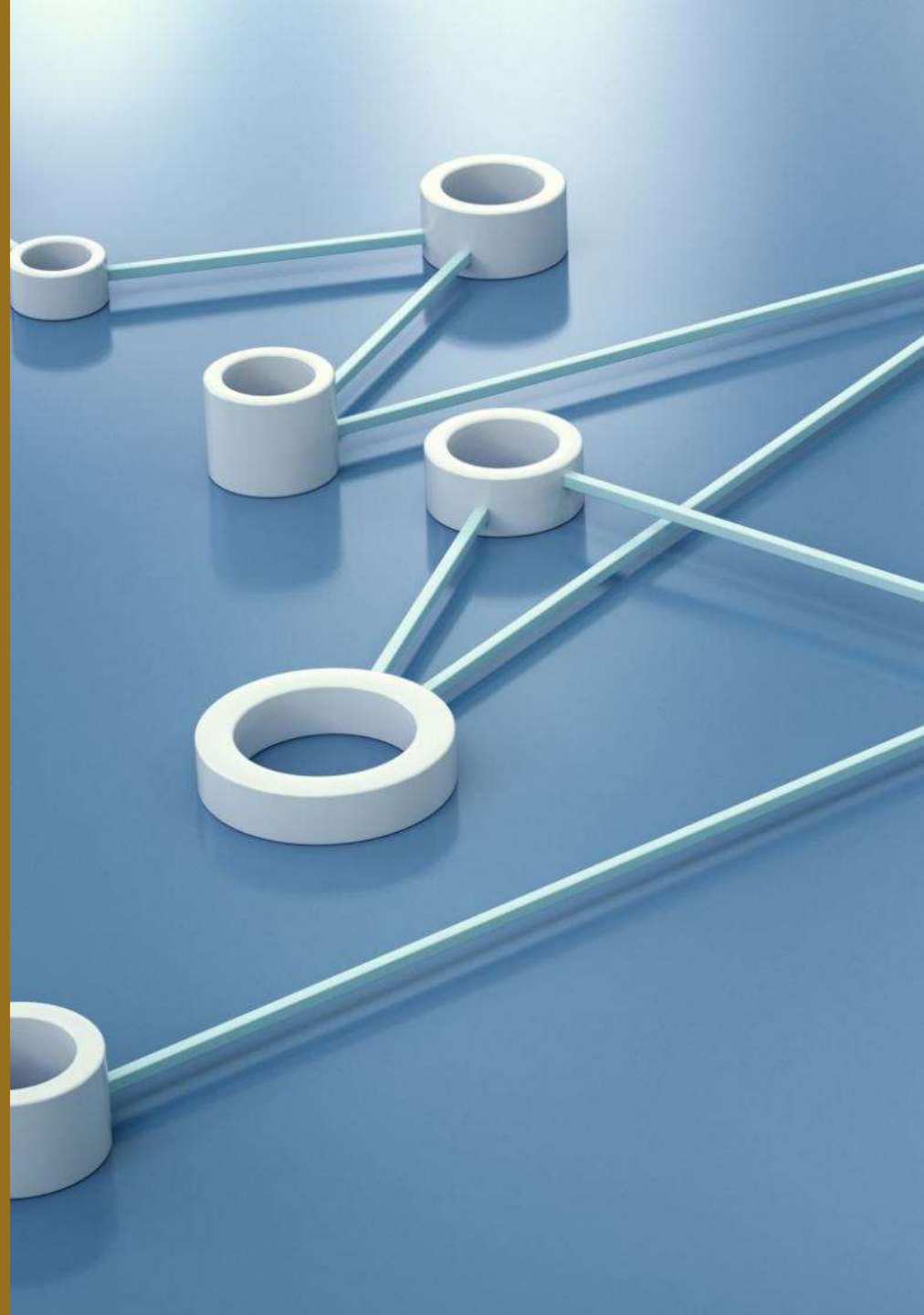
Fig. 1. Schematic view of the new drug development process in the pharmaceutical industry (*source*: Mathieu, 2005, p. 162, modified with data from PhRMA, 2007). *: 20–100 volunteers).

Large firms and innovation

- Large firms that are succesful innovators
  - Have many resources for innovation
  - Often encourage and support multiple approaches to fostering innovation.
  - Focus on continuous improvements
  - They may have several projects under development at any one time, recognising that some will succeed, and some will fail.
- Learning and investigation among employees are valued within such organisations, and there is a high degree of interactive or mutual learning among people.
- Some organisations have 'skunk works' i.e. sub-units that have a high degree of autonomy
- However, there is also bureaucracy, NIH syndrome, focus on exploitation, fear of failure
- Sometimes, Innovations require high-level sponsors who have the power to keep the project alive. Potential sponsors need to see commitment and the ability to follow through.

Not invented here syndrom

- Organisations must increasingly look for external sources of ideas, knowledge and technology to complement their internal capabilities. To fail to do this puts the organisation at a significant competitive disadvantage, leading to eventual failure if the situation persists.
- External sourcing in the innovation process can be difficult to successfully implement, and the key barriers are 'Not-Invented-Here Syndrome' and a lack of 'Absorptive Capacity'.
- *The tendency of a project group of stable composition to believe it possesses a monopoly of knowledge in its field which leads it to reject ideas from outsiders, to the likely detriment of its performance.* (Katz & Allan)



BUSINESS SCHOOL
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Investments in R&D and firm size

Enterprises expenses for own R&D by time, expenses for own R&D and industry - size class - region etc.

	Below 10 employees f.t.e.	10-49 employees f.t.e.	50-249 employees f.t.e.	Above 249 employees f.t.e.	DENMARK TOTAL
2021					
Expenses for own R&D total	990	3 157	7 846	32 609	44 602
Operating costs, wages	529	2 392	6 008	20 459	29 388
Operating costs, other	353	559	1 511	9 335	11 759
Investment costs, buildings	13	72	90	594	768
Investment costs, instruments etc.	95	134	237	2 221	2 686

R&D full-time employees in enterprises by time, industry - size class - region etc. and type of staff

	Total full-time employees	Researchers	Other R&D-personel, including technicians
2021			
SIZE GROUPS (no. of f.t.e.)	38 519	26 308	12 210
Below 10 employees f.t.e.	841	689	152
10-49 employees f.t.e.	3 926	3 113	813
50-249 employees f.t.e.	7 994	5 835	2 160
Above 249 employees f.t.e.	25 757	16 671	9 087
DENMARK TOTAL	38 519	26 308	12 210

Rank	Company	Year	Industry	R&D (€ million)	R&D one-year growth (%)	R&D intensity (%)	Employees
15	NOVO NORDISK	2022	Pharmaceuticals & Biotechnology	2926,224347	33,6	12,3	55185
70	GENMAB	2022	Pharmaceuticals & Biotechnology	588,310809	35,5	30,0	1660
72	DANSKE BANK	2022	Banks	550,255504	-12,3	8,9	21022
77	VESTAS WIND SYSTEMS	2022	Alternative Energy	514	15,8	3,5	28438
82	H LUNDBECK	2022	Pharmaceuticals & Biotechnology	494,450021	-1,4	20,2	5450
83	DANFOSS	2022	Industrial Engineering	481	33,6	4,7	41324
97	ASCENDIS PHARMA	2022	Pharmaceuticals & Biotechnology	368,732	29,0	720,5	797
112	LEO PHARMA	2022	Pharmaceuticals & Biotechnology	321,116848	-15,9	22,4	5252
144	GN STORE NORD	2022	Technology Hardware & Equipment	239,089512	12,1	9,5	7891
145	NOVOZYMES	2022	Pharmaceuticals & Biotechnology	238,686099	-1,8	10,1	6781
147	LEGO	2022	Leisure Goods	231,02125	63,9	2,7	23775
154	GRUNDFOS	2022	Gas, Water & Multiutilities	211,254007	22,6	4,7	20234
155	WILLIAM DEMANT INVEST	2022	Financial Services	210,581651	0,8	6,1	23383
170	T&W MEDICAL	2022	Health Care Equipment & Services	179,639865	17,5	7,6	12111
190	ARLA FOODS	2022	Food Producers	162	20,9	1,2	20907
192	BAVARIAN NORDIC	2022	Pharmaceuticals & Biotechnology	158,230596	198,3	37,3	975
237	COLOPLAST	2022	Health Care Equipment & Services	109,190486	14,4	3,6	14572
243	SIMCORP	2022	Software & Computer Services	102,719	17,7	18,3	2245
267	ALK ABELLO	2022	Pharmaceuticals & Biotechnology	88,213003	5,3	14,5	2680
287	ZEALAND PHARMA	2022	Pharmaceuticals & Biotechnology	80,024524	4,8	310,6	196
288	TDC	2022	Fixed Line Telecommunications	80,01027	1,5	3,7	6477
306	CHR HANSEN	2022	Pharmaceuticals & Biotechnology	73,7	1,7	6,1	3693
319	AMBU	2022	Health Care Equipment & Services	69,655999	1,8	11,7	4409
335	SITECORE HOLDING II	2022	Software & Computer Services	61,7	31,3		
344	ROCKWOOL	2022	Construction & Materials	59	18,0	1,5	12197
396	3SHAPE	2022	Real Estate Investment & Services	47,231877	54,1	11,4	2022
400	FLSMIDTH	2022	Industrial Engineering	45,720154	15,6	1,6	10977
421	BANG & OLUFSEN	2022	Leisure Goods	40,07237	-6,0	10,8	997
433	NKT	2022	Electronic & Electrical Equipment	37,3	-22,5	1,7	4471
468	DSV	2022	Industrial Transportation	32,004108	-3,6	0,1	76283
479	NILFISK HOLDING	2022	Household Goods & Home Construction	30,8	20,3	2,9	4655

Innovation Dynamics: Small vs. Large Firms

- Resource Allocation:
  - Small and large firms allocate resources differently for innovation endeavors, influenced by financial capabilities, access to talent, and risk tolerance.
- Organization of Innovation Processes:
  - the organizational structure and processes for innovation can vary between small and large firms, impacting agility, decision-making, and adaptability.
- Search for New Ideas:
  - Small firms **may** employ more agile and informal methods in their quest for new ideas, while large firms **may** utilize structured processes and dedicated research teams.
- Collaboration Patterns:
  - Collaboration patterns differ, with small firms often engaging in close-knit partnerships and large firms leveraging extensive networks and strategic alliances.
- Outcome Variability:
  - The outcomes of innovation efforts may vary, influenced by factors such as market dynamics, resource allocation, and organizational culture.
- Types of Innovation Outcomes:
  - Both small and large firms can produce various types of innovation outcomes, including incremental improvements, disruptive breakthroughs, and novel business models.
- Managerial vs. Entrepreneurial Approach:
  - Managers in large firms may prioritize efficiency and risk management, while entrepreneurs in small firms often embrace experimentation and agility to drive innovation.

Creating the Innovative Organisation 1

- 1. market orientation;
- 2. innovative leadership;
- 3. non-linear strategic planning;
- 4. ambidextrous structure;
- 5. innovation focused culture.

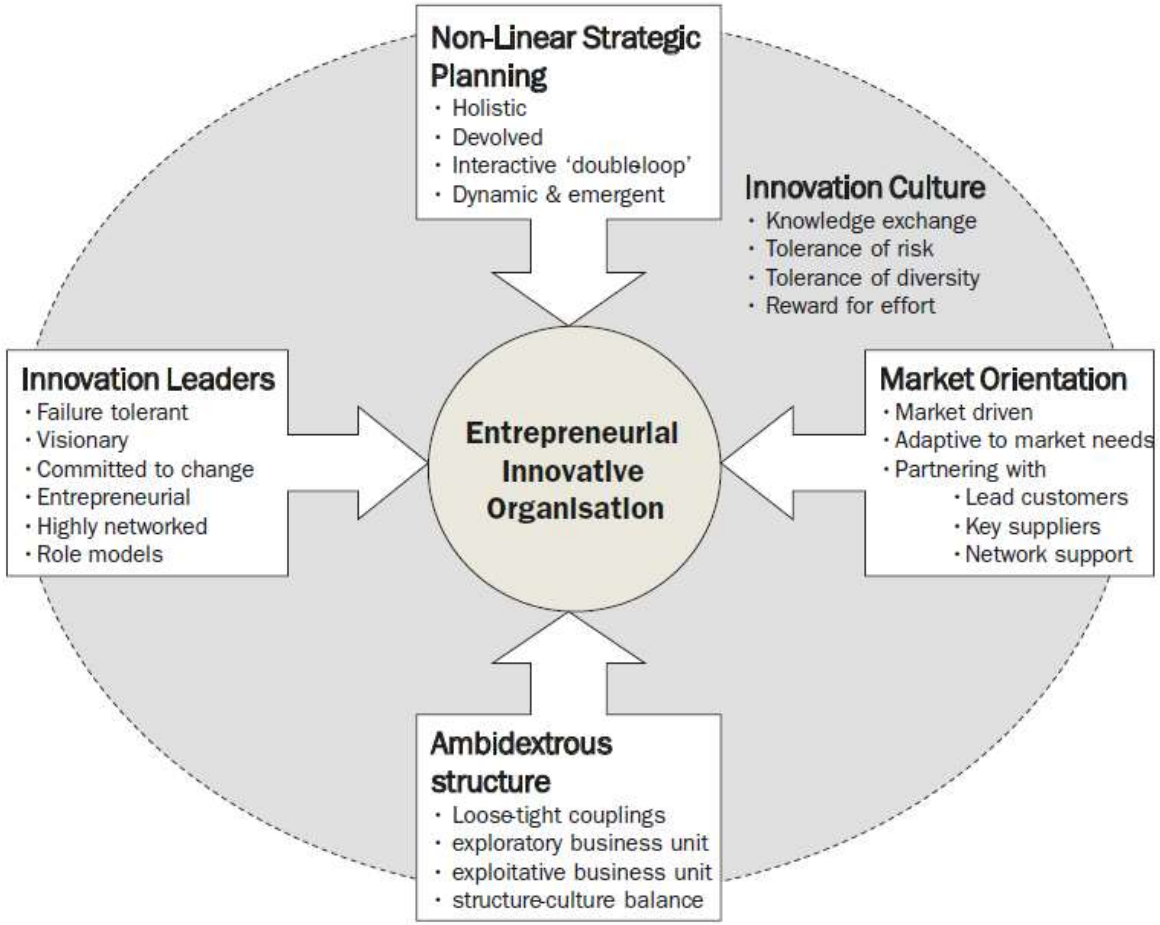

Fig. 4.4 A model of entrepreneurial innovative organisations. (Source: Mazzarol 2004)

Creating the Innovative Organisation 2

- Market orientation
  - firm to closely monitor and respond to the needs of customers
  - Establish strategic partnerships with **lead customers** and **key suppliers** as well as universities or other knowledge organisations
- Innovative leadership
  - Managing innovation within an organisation requires leadership to set strategic directions, to motivate and empower employees, and to guide activity in desired directions
  - capturing people's attention and focusing their efforts on innovation
  - get innovative ideas from people's heads and into action
- non-linear strategic planning
  - Innovation management should be viewed as a strategic process with a formal strategic innovation plan
  - managers also need to learn how to identify barriers to innovation
  - involves a process of the continuous screening of opportunities, the weeding out of less promising options selection, and prompt exploitation

Creating the Innovative Organisation 3

- ambidextrous structure
  - Exploration and exploitation in two distinct units
  - a common impediment is the organisational structure. By its nature innovation involved new combinations of skills, resources and technologies that challenge the existing system
  - Organisational structures that are better suited to radical innovations usually have a smaller, flatter management structure and cross-functional teams that operate with greater autonomy than is usual for the larger organisation.
- innovation focused culture.
  - Innovative behaviour among employees has been found to be positively associated with the level of support for innovation engendered within the organisation's culture.
  - Senior management will need to be more tolerant of failure
  - Investment in human resources via training and skill development

Innovation in small and large firms

- Opportunity recognition
  - Manager versus management
  - Formalised R&D processes
- Resource accumulation
  - R&D spending
  - Absorptive capacity
  - Networking practices
  - Collaboration methods and extent
  - Open innovation strategies
- Capacity building for innovation capture
  - Strategy
  - IPR protection
  - Sales

What have you learned today

- Innovation as an interactive process with different stages
- Innovation in small and large firms
- Small firms have less resources and rely more on flexibility and collaboration
- Large firms have many resources and have the possibility to be more strategic in their innovation strategy
- Role of absorptive capacity
- Open innovation
- Role of management and organisational structure